Qiskit in Classrooms for the Qiskit Fall Fest 2026

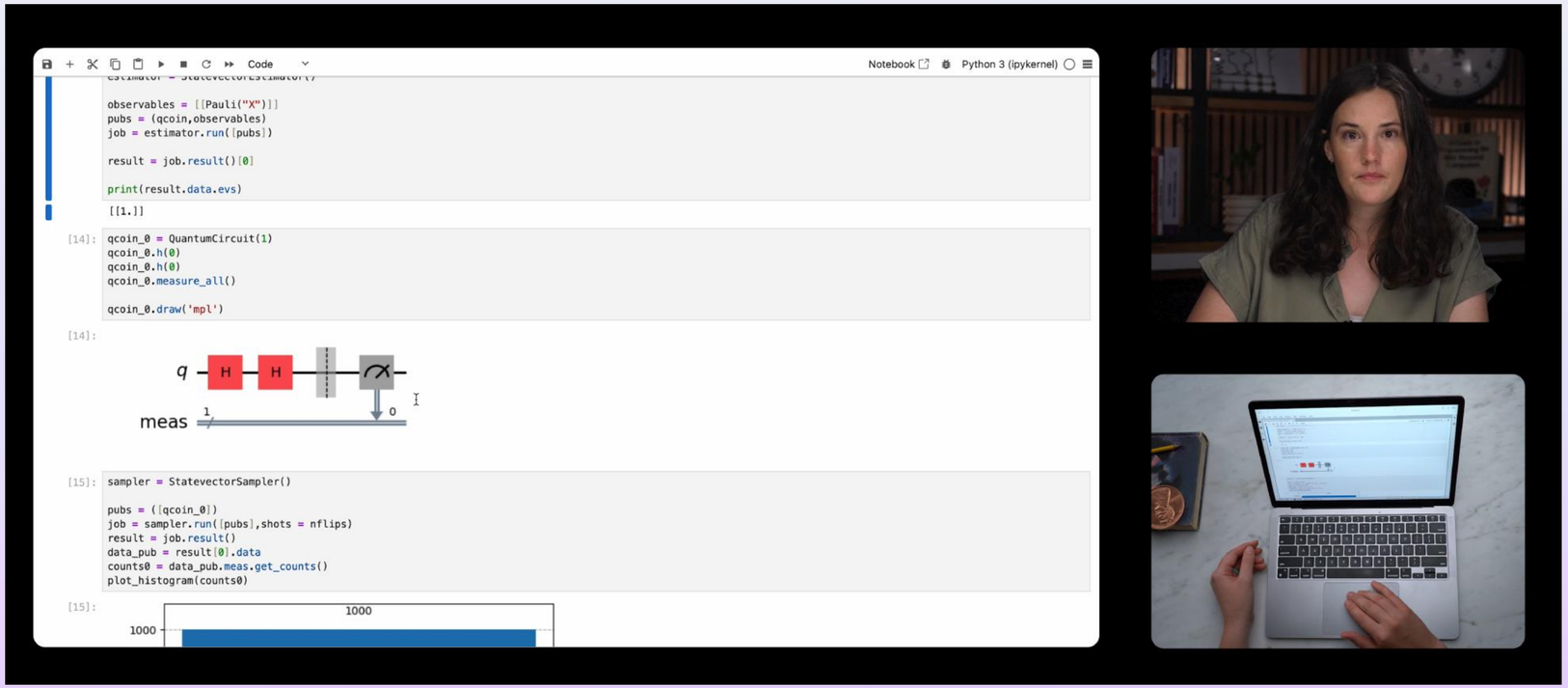
Sophy Shin

Quantum Developer Advocate

IBM **Quantum** Learning

Qiskit classroom modules

IBM **Quantum** Learning





Katie McCormick



Christopher Porter



Sophy shin



Olivia Lanes

Qiskit classroom modules

Purpose:

- Leverage quantum computers for experimental investigation of quantum phenomena.
- Get the future quantum workforce using real quantum computers early
- Target audience: most likely future quantum practitioners

What the modules are:

- Modules are ~1-2 hour-long, Jupyter notebooks on key concepts from a quantum-computing related course.
- Text and Qiskit code, check-in questions and coding activities for the learner to check their understanding.
- Designed to be easily plugged into courses with minimal effort from the instructor.
- Optional videos paired with each module

```
18  estimator = Estimator(mode = backend)
19  job = estimator.run([[qc_isa,obs_isa]])
20  res = job.result()
21
22  # Run the job on the Aer simulator with noise model from real backend
23  #job = estimator_sim.run([[qc_isa,obs_isa]])
24  #res=job.result()
25
26  #Step 4: Return the result in classical form, and analyze.
27
28  print(res[0].data.evs)
```

✓ 21.3s Python

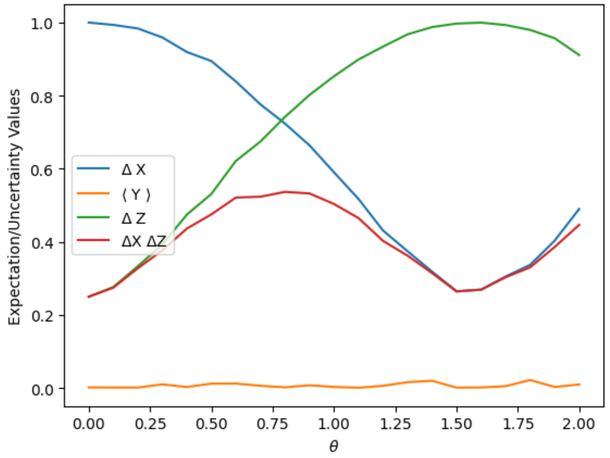
```
-0.02906350914962325
```

According to our equation above,

$$(\Delta X)^2 = 1 - \langle X \rangle^2 = 1 - (0.0107)^2 \rightarrow \Delta X = 0.999...$$

Let's stick with that same state, but find the expectation value of Z, now:

Uncertainty relations



The 12 modules at a glance

Foundation

- Get Started with Qiskit

Quantum Mechanics

- Stern-Gerlach Experiment
- Quantum Coins (Superposition)
- Bell’s Inequality
- Uncertainty

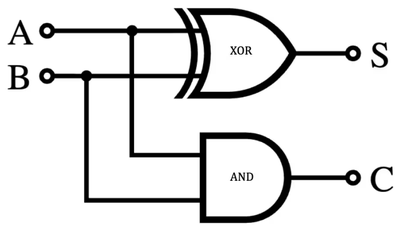
CS / CSE / Algorithm/ Chem

- Quantum Teleportation
- Quantum Key Distribution
- Deutsch-Jozsa
- Grover’s Algorithm
- Shor’s Algorithm
- QFT & Phase Estimation
- Variational Quantum Eigensolver

Foundation

Get Started with Qiskit – the on-ramp to quantum computing

Module 01



- Rebuilds the classical half-adder as a qubit circuit

Concepts covered

- Classical bits & gates
- Qubits & quantum gates
- Measurement
- Quantum circuits
- Superposition / entanglement
- Qiskit conventions (little-endian) / Qiskit Patterns workflow

What students do

- Build the classical half-adder, then adapt it to run on qubits.

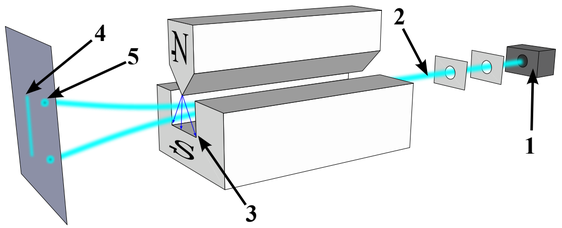
Real-QPU time

- ~2 s

Quantum Mechanics

Stern-Gerlach Experiment
– a module on spin quantization

Module 01



- Uses real qubits to reproduce original Stern-Gerlach results and extensions

Concepts covered

- Spin quantization ("up" / "down")
- Spin-½ measurement outcomes
- Repeated measurements & noise
- Sequential / different-axis measurements
- State vectors & matrix operators for spin

Prerequisite

- Get Started module

What students do

- Ex1: single measurement
- Ex2: many particles
- Ex3: “random spins in an oven”
- Ex4: repeated measurements
- Ex5: different observables.

Real-QPU time

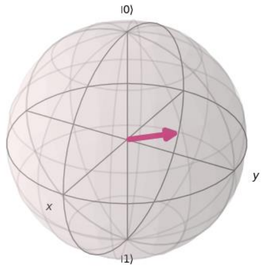
- ~2 s

Quantum Mechanics

Superposition– a module on superposition and interference

Module 02





- Compares measurement of a qubit in equal superposition to a coin flip and explores limits of that analogy.

Concepts covered

- Qubit vs. classical coin flip
- Superposition & measurement probability
- Phase coherence
- Constructive / destructive interference
- Bloch sphere (θ, φ)
- Single-qubit gates as rotations

Prerequisite

- Get Started module

What students do

- 3D “quantum coin” experiment
- the √NOT gate
- visualize the state as a Bloch vector.

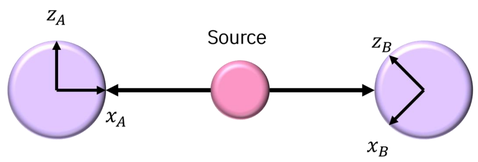
Real-QPU time

- ~47 s

Quantum Mechanics

Bell's Inequality – a module on entanglement and disproving local realism

Module 03



- Shows how quantum mechanics violates Bell's inequality using a real quantum computer.

Concepts covered

- Entanglement
- Hidden variables / local realism
- Bell (CHSH) inequality
- Probabilistic QM vs. classical correlation
- Real-world scenario → quantum circuits

Prerequisite

- Get Started module

What students do

- Frame Einstein's (hidden-variable) vs Born's (QM) predictions
- test the inequality on hardware.

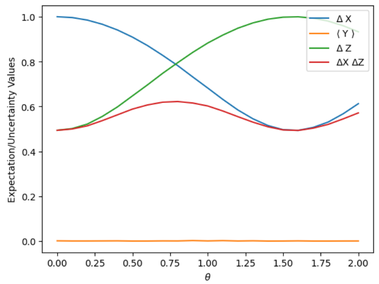
Real-QPU time

- ~12 s

Quantum Mechanics

Quantum Uncertainty – a module on uncertainty relations

Module 04



- Checks the generalized uncertainty relations of Pauli observables with a qubit

Concepts covered

- Uncertainty relations (position–momentum, spin)
- Non-commuting operators / Pauli matrices
- Commutators
- Uncertainty bound $\Delta A \cdot \Delta B \geq \frac{1}{2}|\langle[A,B]\rangle|$
- Hardware verification of operator intuition

Prerequisite

- Get Started module

What students do

- Measure uncertainties
- Test the uncertainty relation.

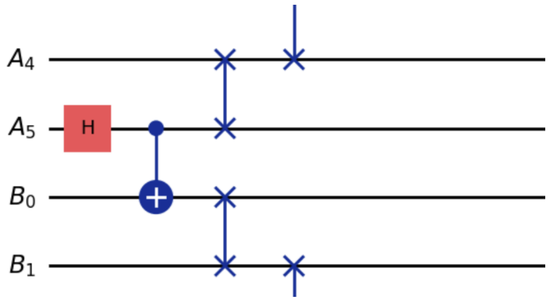
Real-QPU time

- ~8min. plan lab time / use simulator fallback

CS/CSE/Chem

Teleportation – a module on teleporting information

Module 01



- Moves information on a real QPU from one set of qubits to another.

Concepts covered

- Entanglement & measurement
- Bell pairs
- State transfer (teleportation)
- Classical communication → no faster-than-light
- Noise & errors on real QPUs

Prerequisite

- Get Started module

What students do

- Ex1: basic teleportation
- Ex2: teleport across the processor.

Real-QPU time

- ~14 s

CS/CSE/Chem

QKD – a module on the BB84 protocol for quantum key distribution

Module 02



- Walks through the BB84 protocol for securely sharing a decryption key using quantum states.

Concepts covered

- No-cloning
- One-time pad
- Random bases & basis reconciliation
- Key sifting
- Eavesdropper detection

Prerequisite

- Get Started module

What students do

- BB84 steps
- Ex1: no eavesdropper
- Ex2: eavesdropper
- Ex3: compare on hardware.

Real-QPU time

- ~5 s

CS/CSE/Chem

Deutsch-Jozsa – a module on the DJ algorithm for balanced/constant functions

Module 03

x	$f_1(x)$	$f_2(x)$	$f_3(x)$	$f_4(x)$
0	0	0	1	1
1	0	1	0	1

- Implements the DJ algorithm to check function types and discusses extensions.

Concepts covered

- Constant vs. balanced functions
- Quantum parallelism
- Phase kickback / Interference
- Polynomial speedup / query model
- Bernstein-Vazirani (hidden string)

Prerequisite

- Get Started module

What students do

- Quantum parallelism
- Deutsch
- Deutsch-Jozsa
- Bernstein-Vazirani.

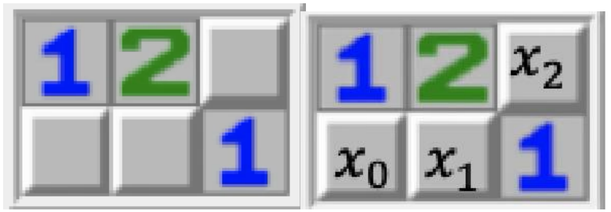
Real-QPU time

- ~4 s

CS/CSE/Chem

Grover's – a module on unstructured search

Module 04



- Amplifies target states for an O(√N) unstructured search; culminates in solving a Minesweeper grid.

Concepts covered

- Unstructured search
- Quadratic speedup O(√N)
- Oracle & Grover operator
- Amplitude amplification & iteration count
- Iterations vs. hardware noise
- Oracle from constraints (Minesweeper)

Prerequisite

- Get Started module
- Deutsch-Jozsa (oracles)

What students do

- 3 activities:
  - single target
  - accurate query workflow
  - Minesweeper grid.

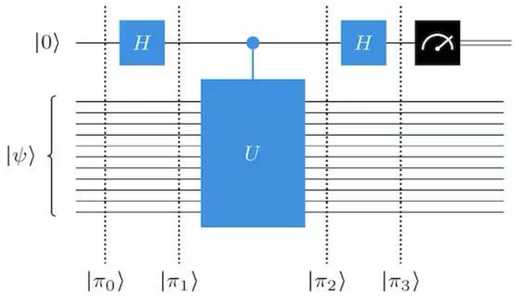
Real-QPU time

- ~12 s

CS/CSE/Chem

QFT & QPE – a module on the quantum Fourier transform

Module 05



- Builds the QFT and quantum phase estimation — reusable subroutines behind Shor's and many algorithms.

Concepts covered

- Discrete Fourier transform
- Quantum Fourier transform
- Fourier basis (basis transformation)
- Phase kickback
- Quantum phase estimation
- Reusable algorithm subroutine

Prerequisite

- Get Started module

What students do

- Fourier-basis intuition
- QFTs
- QPE with more precision.

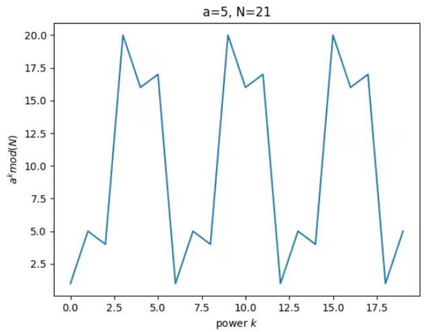
Real-QPU time

- ~13 s

CS/CSE/Chem

Shor's – a module on factoring via quantum order finding

Module 06



- Reduces factoring to quantum order finding with phase estimation; the algorithm behind "quantum breaks RSA."

Concepts covered

- Integer factoring
- Cryptography & RSA relevance
- Modular arithmetic
- Order finding
- Quantum phase estimation
- Classical–quantum hybrid workflow

Prerequisite

- Get Started module
- QFT/QPE

What students do

- Modular arithmetic
- Order finding
- Map / Optimize / Execute / Post-process.

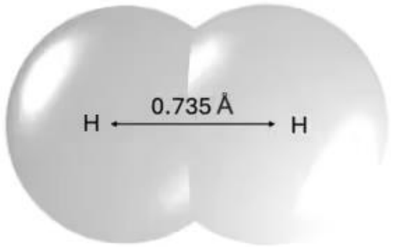
Real-QPU time

- ~3 s

CS/CSE/Chem

VQE – a module on ground-state energies in chemistry

Module 07



- A variational quantum-classical algorithm for ground-state energies in chemistry.

Concepts covered

- Variational principle
- Hamiltonian → qubit mapping
- Ansatz (parameterized circuit)
- Hybrid quantum–classical loop
- Ground-state energy
- Reaction energy

Prerequisite

- Get Started module

What students do

- Ground-state energy of H atom
- H_2 molecule
- Reaction energy of H + H = H_2.

Real-QPU time

- ~8 min

How a host uses the modules

Every module ships with automated grading (qc_grader · fallfest_2026 · 130 exercises)

1. Live / on-site

- Work modules together at a booth or session
- Grader = instant check-in feedback (🎉 / 😕)

2. Homework

- Assign a path with a deadline
- Grader autograde; check_progress()

3. Self-paced challenge

- Pick a path, race the clock

How a host uses the modules

Examples:

- *Weekend event*
- *40 participants*
- *4 mentors*
- *1 organizer*

Tips:

Mentors and organizers can also watch each module's companion videos, then teach the module live themselves.

Open-ended discussion questions aren't auto-graded — use them for group discussion

At the end, you can run a live quiz to check participants' understanding.

Participant

- Pick a path that fits your interest
- Solve → run grade_<mod>_exN(answer)
- See 🎉/😕 instantly; retry freely
- check_progress() tracks your completion
- Queue busy? switch to simulator cell

Mentor

- Pre-run the sticky exercises before the event
- Debug tips: data type / shape, qubit order (little-endian q_0)
- Steer to simulator when the QPU queue is long
- 5-min mini-talk when many hit the same wall

Organizer

- Announce completion rules up front
- Aggregate check_progress()
- Count points and award them

Example tracks

Get Started with Qiskit — bits/gates → qubits, circuits (prerequisite for every path)

Physics Track

Stern-Gerlach
↓ *spin quantization*
Quantum Coins
↓ *superposition · Bloch*
Uncertainty
↓ *non-commuting operators*
Bell's Inequality (capstone)

CS Track

Teleportation
↓ *entanglement · measurement*
QKD (BB84)
↓ *no-cloning*
Deutsch-Jozsa
↓ *phase kickback*
Grover (capstone: Minesweeper)

Algorithm track

Deutsch-Jozsa
↓ *oracles · kickback*
QFT & Phase Estimation
↓ *reusable subroutine*
Shor's Algorithm
↓ *hybrid workflow*
VQE (chemistry · hybrid)